

Board sheet

Why you are holding this. Most of today is on the board, and copying a derivation is not the same as following one. The scaffolding is printed; **the lines are blank on purpose**. Fill them in as they go up, and spend the time you save on noticing *why* each step follows. Nothing here is collected. Everything here is on PS1.

1 • Where mass action comes from 8–13 min

A and B react only when they meet. In a well-mixed box the number of A–B encounters per second is proportional to how many of each there are.

Two A and three B. Distinct pairs that can meet: _____

Three A and three B: _____

So the rate law is

What does k absorb? (list at least three things)

A stoichiometric coefficient enters the **flux as a power** and the **balance as a multiplier**. For $P + P \rightarrow P_2$ the flux goes as _____, not _____.

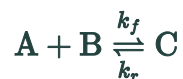
2 • Is mass action allowed in an *E. coli*? 13–17 min

Three assumptions, three numbers. **Given:** $L = 1 \mu\text{m}$, $D = 7.7 \mu\text{m}^2/\text{s}$, $k_{\text{on}} = 10^8 \text{ M}^{-1}\text{s}^{-1}$, $C = 1 \text{ nM}$.

	TEST	NUMBER	VERDICT
well mixed	$\text{Da} = \tau_{\text{mix}}/\tau_{\text{rxn}}$	$\tau_{\text{mix}} \approx$ $\tau_{\text{rxn}} \approx$ $\text{Da} \approx$	
dilute	activity $\propto$ concentration?	20% protein by weight	
many	fluctuations go as $1/\sqrt{N}$	$1 \text{ nM} \approx$ molecules, so $N =$	

The three answers do not agree. Which one fails, and is it the one you expected?

3 · The long way 17–21 min



$$\frac{d[A]}{dt} =$$

$$\frac{d[B]}{dt} =$$

$$\frac{d[C]}{dt} =$$

How many **distinct fluxes** are there in those six terms? _____

4 · Two fluxes, one matrix 21–26 min

$$v_1 =$$

$$v_2 =$$

S_{ij} = net molecules of species i produced by reaction j .

	v_1 (FORWARD)	v_2 (REVERSE)
A		
B		
C		

$$\frac{d\mathbf{x}}{dt} =$$

$\mathbf{S}$ is _____ — it comes off the reaction list alone and knows nothing about rate constants, concentrations or time.

$\mathbf{v}$ is _____ — every parameter and all of the biology.

5 · Two null spaces 26–33 min

LEFT. If $\mathbf{w}^T \mathbf{S} = \mathbf{0}$ then $\frac{d}{dt}(\mathbf{w}^T \mathbf{x}) =$ _____ for any $\mathbf{v}$ — so $\mathbf{w}^T \mathbf{x}$ is conserved.

Find two for $A + B \rightleftharpoons C$:

$\mathbf{w} =$ _____ giving _____ $\mathbf{w} =$ _____ giving _____

$\text{rank } \mathbf{S} =$ _____, so $\text{dim}(\text{left null}) = n - \text{rank } \mathbf{S} =$ _____

RIGHT. $\mathbf{S}\mathbf{v} = \mathbf{0}$ asks: which patterns of reaction rates leave every concentration unchanged? Here that gives _____, so $\text{dim}(\text{right null}) = r - \text{rank } \mathbf{S} =$ _____

The cascade, for contrast. $\emptyset \xrightarrow{\alpha} m$, $m \xrightarrow{\gamma_m} \emptyset$, $m \xrightarrow{k_p} m + p$, $p \xrightarrow{\gamma_p} \emptyset$

S is _____ $\times$ _____ with rank _____. Conservation laws: _____. Flux modes: _____.

Why no conservation laws at all?

Flux balance analysis, in session 20, is the right null space plus bounds on $\mathbf{v}$ plus an objective. That is all it is.

6 • The steady state, without a computer 59–63 min

$$\frac{dm}{dt} = \alpha - \gamma_m m = 0 \implies m^* =$$

$$\frac{dp}{dt} = k_p m^* - \gamma_p p = 0 \implies p^* =$$

With $\alpha = 10$, $\gamma_m = 0.5$, $k_p = 4$, $\gamma_p = 0.05$: $m^* =$ _____ $p^* =$ _____

The move worth naming: a cascade solves _____ — m^* does not contain p , so solve it alone and substitute.

7 • Integrate it exactly 63–67 min

$$m(t) = m^* (1 - e^{-\gamma_m t}) \implies \frac{dp}{dt} + \gamma_p p = k_p m(t)$$

Multiply by the integrating factor _____ :

$$\frac{d}{dt} \left[\right] =$$

$$p(t) = p^* \left[1 - \right]$$

Two checks, both worth ten seconds. At $t = 0$: _____. As $t \rightarrow \infty$: _____.

Two exponentials, not one. The protein does not simply relax at γ_p — it also carries the mRNA's rise.

8 • Four parameters become one 70–74 min

Scale by the answer: $\tau = \gamma_p t$, $\mu = m/m^*$, $\pi = p/p^*$.

$$\frac{d\mu}{d\tau} = \quad \quad \quad \frac{d\pi}{d\tau} =$$

$\varepsilon =$

and ours is

That one number is the **only** parameter left. Two cascades with wildly different rate constants and the same ε give the same curve after rescaling.

Setting $\varepsilon = 0$ says $\mu = 1$ instantly. That is the **quasi-steady-state approximation**, and Tuesday is about what it costs: the largest error in p/p^* turns out to be about _____.

Take away

1. Every deterministic model in this course is $\dot{\mathbf{x}} = \mathbf{S}\mathbf{v}(\mathbf{x})$; structure and kinetics separate.
2. A species on both sides of a reaction has **zero** net stoichiometry — translation is catalytic in the message.
3. Find the conserved quantity and check it. The cheapest test you will ever write.